Cybersecurity Fundamentals & SOC operations

TechBiz Academy | SOC Analyst Internship | Week 1 – Task 1

01 – CIA Triad

CIA Triad are the 3 core pillars of cybersecurity; **C**onfidentiality means the data/information is only accessible to the ones who are authorized to access it. **I**ntegrity means the data is not altered and is accurate, for example when downloading an ISO file, I would check the checksum of the ISO to verify the file I downloaded and the file on the official website is the same to ensure its integrity. **A**vailability, data/information should be available to the authorized personnel when it is necessary. Collectively these principles are in focus when creating a security plan or protecting data and information.

02 – Threat, Vulnerability and Risk

- **Threats** are the actors who try to use the vulnerability/ weakness in a system to exploit them.
- **Vulnerability** is the weakness in the system like an unpatched software or unsupported system with EOL which the threats might be able to exploit.
- **Risk** is the likelihood or the impact of a threat exploiting a vulnerability.

For example: A website which has transfers username and password in clear text could be vulnerability, an attacker capturing packets of the network could be a threat and the attacker using the clear text password to login to the system and exporting the records could be a risk.

03 – Definition and Purpose of SOC

Security Operation Center (SOC) is a dedicated team consisting of SOC analysts, Security Engineers, Incident Responders and Threat hunters who collectively work together in improving the cybersecurity posture of a company by preventing, detecting, and responding to threats.

The purpose of a SOC is to prevent, react and improve security of company with Threat Detection by continuous monitoring of organization's entire environment with tools like SIEM, SOAR, and XDR. Incident response by quickly taking action to limit damage of the attack and ensure the apps tools and processes are compliant with regulations and standards.

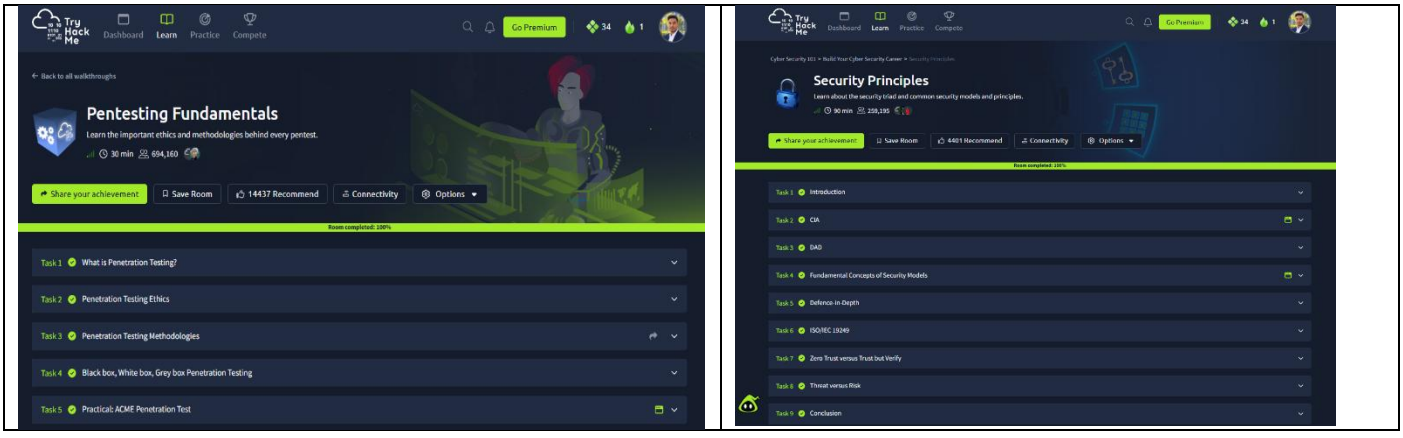
Role	Responsibilities
L1 Analyst	Monitor Alerts, perform triage, collect evidence, prioritize and escalate activities.
L2 Analyst	Investigate escalated alerts, correlate logs, assess impact, coordinate containment.
L3 Analyst	Investigate complex threats perform advanced analysis, support difficult escalations.

04 – Red team and Blue team

Red team	**Blue Team**
Offensive Security	Defensive security
Simulates authorized attacks	Detects and responds to attacks
Tests security controls	Operates and improves security controls
Identifies weakness in defense	Investigates suspicious activity
Reports findings and recommendations	Documents incidents and improves defense.

05 – Practical Learning and Evidence

TryHackMe room completion



Notes link:

https://app.notion.com/p/Week-1-Task-1-notes-414f4771e2948045ba45ea51ac6bb69c?source=copy_link